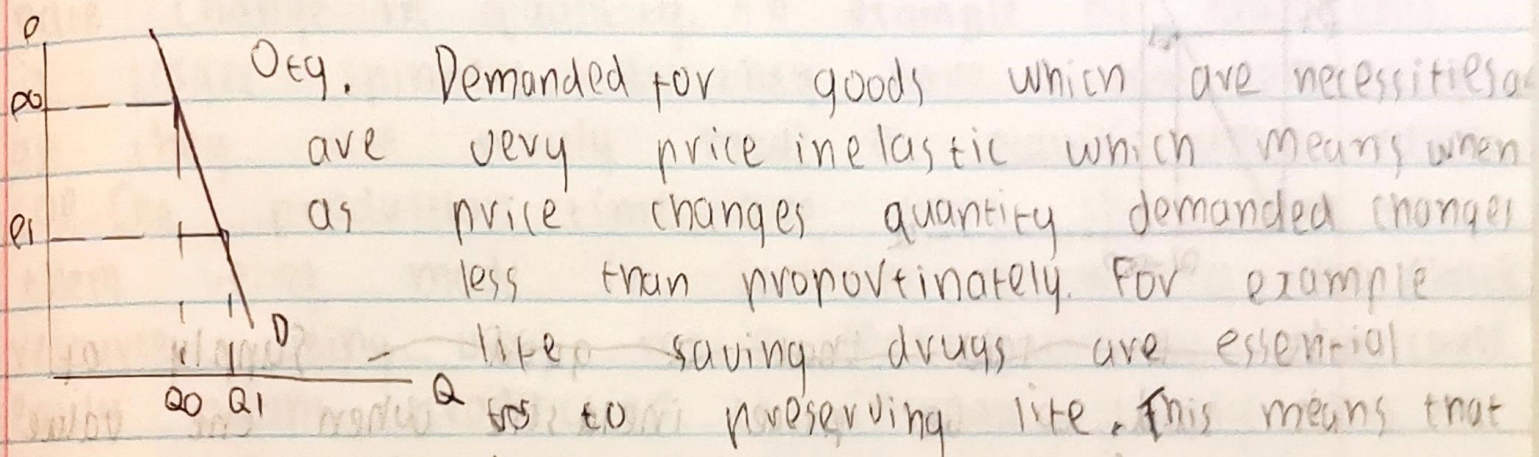
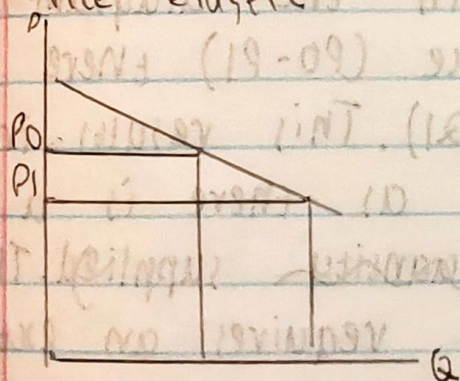


PED — is a measure of the response or reaction in quantity demanded to a change in price of a product. This can be calculated by dividing Percentage change in quantity demanded by Percent change in price. Elasticity will generally be negative due to the law of ~~supply~~ demand. It is said that any PED that is less than 1 is said to be inelastic while any value from $1 - \infty$ is price elastic. When price is elastic a change in price results in a more than proportionate change in quantity. This is contrasted with an inelastic demand where a change in price results in a less than proportionate change in price.

Price Inelastic:



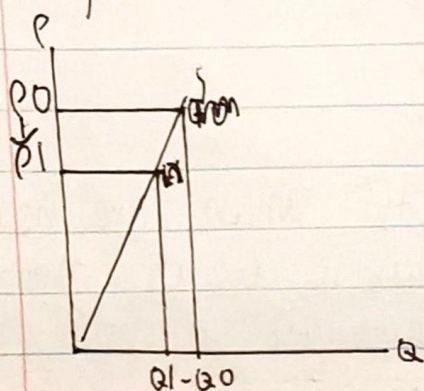
despite a change in price of these drugs the quantity demanded will not be drastically effected.



Goods which are necessities are very [^]inelastic as despite what the good is priced at consumers demand will stay similar. For example life saving drugs are essential to preserving life so firms can set any price without demand being drastically effected.

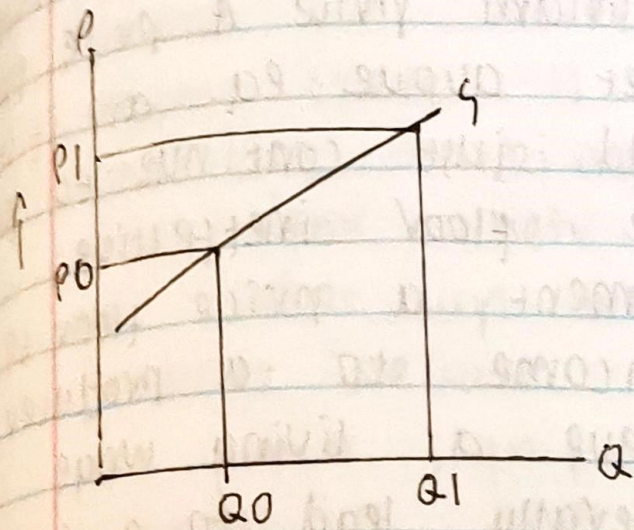
PES- Elasticity of supply is a measure of the reaction or response of a firm in quantity supplied to a market following a change in price of a product. It is the ratio of the percentage change in quantity supplied and the percentage change in the price of a product. The sign of PES is positive as it shows that a rise in price causes a rise in quantity.

If there was a ~~change~~ 5% fall in the supply of gold following a 10% fall in price the PES would be 0.5. This shows the responsiveness of quantity to changes in price.



~~It is said that a good~~ Supply of a good is said to be price inelastic when the value is less than one. This can be seen in the diagram as even with a large change in price ($P_0 - P_1$) there is a small change in quantity ($Q_0 - Q_1$). This results in a movement along the supply curve as there is a less than proportionate change in quantity supplied. This can be seen in crude oil as it requires an extensive extraction process which would increase cost of production and price. Suppliers would require time to extract the resource ~~due to~~ ^{via} drilling and exploration which renders crude oil less responsive to price changes.

Supply of a good is said to be price elastic when the value is more than 1 but less than ∞



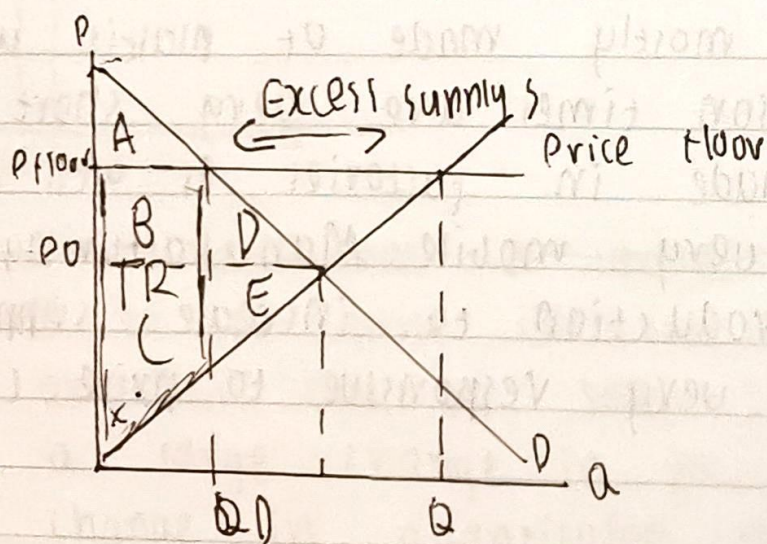
This idea can be seen in the diagram as when P_0 rises to P_1 it creates a more than proportionate change in quantity. An example of ~~rather~~ this is fidget spinners as they have a low complexity as they are mostly made of plastic which reduces COP. The production times are very short due to them being made in factories as well as the plastic resources being very mobile. Manufacturing firms could easily adapt production to increase supply thus making crude oil very responsive to price changes.

Price floor-

A price floor is a price that the government has set above the equilibrium price. A price floor is only effective when set above eq. as if it was below, consumers would just continue to pay eq price thus making the floor ineffective.

A government would implement a price floor to guarantee a level of income to a producer to ensure these workers receive a living wage.

Price ceilings will generally lead to a surplus because the min price is above clearance price. Reduce market size because there is a reduction in quantity demanded at higher price. There would also be a DWL as the market is not operating at market equilibrium.



In this diagram at the original price (P_0) in a free market the CS would be the area ABD while PS would be CE. After this price floor is implemented the CS contracts to just the A area while the PS increases to BC. Regions D and E are not included in TS as they are the DWL and represent inefficiency in the market.

In the diagram there is an excess supply of labour or unemployment ($Q_s - Q_d$) at the price floor. Demand for workers is limited to Q_d which means there is a welfare loss at D. Workers who are working do well. Producers have to pay more for their workers. For producers total revenue went up which increases equity and benefits producer welfare. For consumers the price went up which significantly reduced CS as it reduced affordability and thus equity of the product. This reduces welfare for the consumers and the DWL of the occurring. Overall CS saw a contraction while PS saw an expansion. Although this still resulted in a net dead weight loss which results in total welfare going down for society. This is due to a DWL being formed at DE which highlights market inefficiency.

PS ↑

CS ↓

TS ↓

DWL ↑

Gov spend ↑

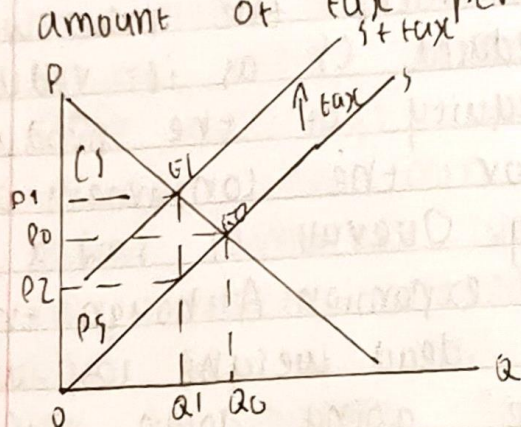
TR ↑

Price ↑

Quant ↓

Q_d ↑

Tax- Indirect taxes are placed on consumption, production or the use of goods and services. They are indirect taxes because tax revenue is raised indirectly through the ability of producers to pass the burden or incidence of the tax to consumers. The type of tax here is a specific tax where there is an absolute amount of tax per unit.



Indirect taxes cause the market supply curve to shift upwards by the amount of tax ($S \rightarrow S+tax$).

If demand is relatively unresponsive to price change (relatively inelastic) the tax will cause the following:
A relatively large increase in price ($P_0 - P_1$) because producers know that if they pass the tax onto consumers it will not lead to a large fall in sales and revenue. Consumers take most of the incidence from the tax but the whole of the tax will only be passed on if demand is perfectly inelastic. It would also cause a relatively small fall in quantity demanded ($Q_0 - Q_1$) because demand is relatively inelastic.

If the demand is relatively responsive to price changes (relatively elastic.) The following will occur:
A relatively small increase in price from $P_1 - P_2$

As producers know that passing a higher price burden on consumers would cause a large fall in sales and revenue. This means that producers take the larger incidence of the tax.

It will also cause a relatively large fall in quantity demanded seen Q_1 to Q_2 because demand is elastic.

When a tax is imposed both consumers and producers lose part of their ~~total~~ surplus to tax revenue. With a good which is very responsive to price changes, producers will bear the brunt of the burden and sacrifice their PS as the ~~demand~~ ~~for~~ total revenue will fall too much with a price rise. Although ~~when~~ when the goods are very price inelastic consumers take the hit as ~~goods~~ they don't care about price.

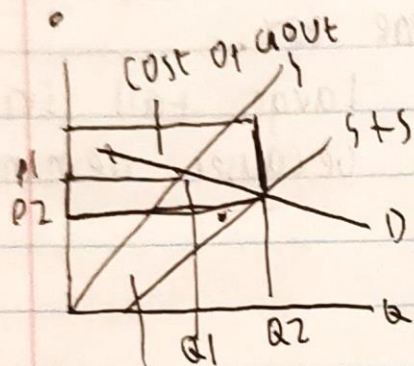
Since both CS AND PS are down the welfare and equity of these groups has gone down in favour of boosting the government's tax revenue. This revenue could be reinvested into the same market to alleviate the dead weight loss that can be seen. This deadweight loss is a lack of efficiency in the market. Although this revenue is often not reinvested into the same market resulting in a large DWL and an inefficient market.

$CS \downarrow$
 $PS \downarrow$
 $TR \downarrow$
 $TW \downarrow$

Price $\uparrow$
DWL $\uparrow$
Tax $\uparrow$

A subsidy is a payment by the government to a firm in order to increase output

• Cause DWL



Prod surplus

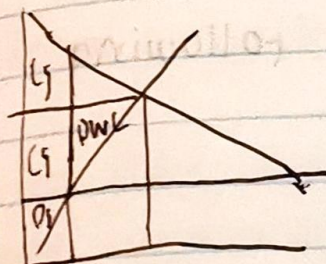
Cost of Govt and taxpayer is higher than gains in producer surplus.

- ↑ Equity for Consumers + Producers
- ↑ Total Surplus
- ↑ Total welfare
- ↑ Total revenue
- ↓ Tax revenue
- ↓ OPP cost

Inelastic = ~~subsidy~~ subsidy

Elastic = No subsidy

Price ceiling



PS ↓
CS ↑

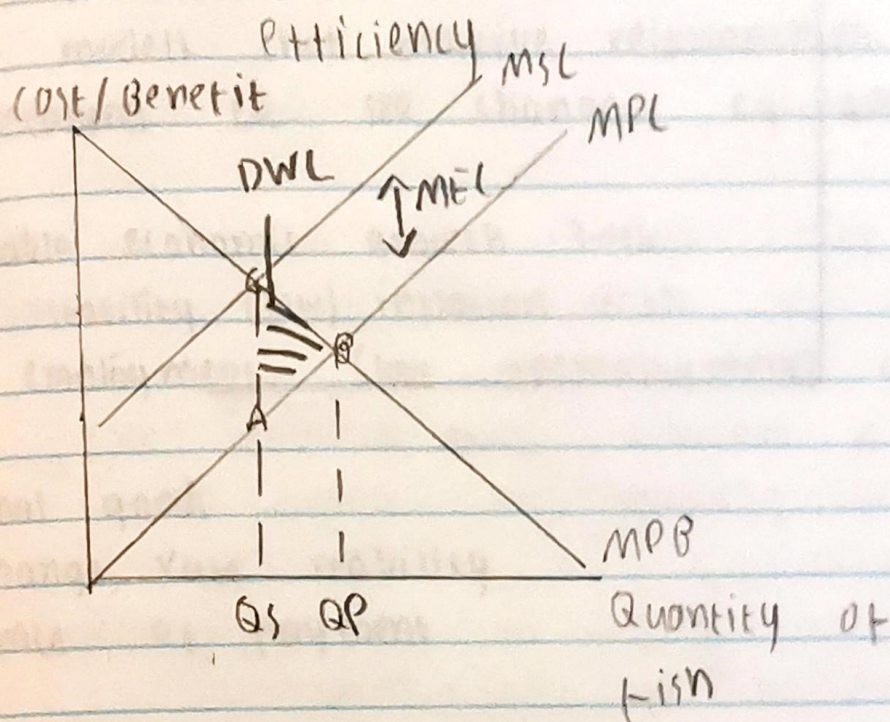
TS ↓
DWL ↑

Price ↓

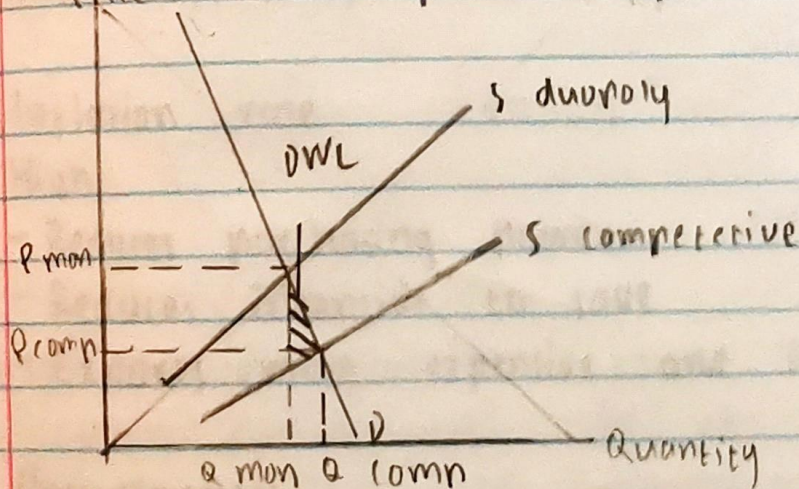
TR ↓

Quant ↓

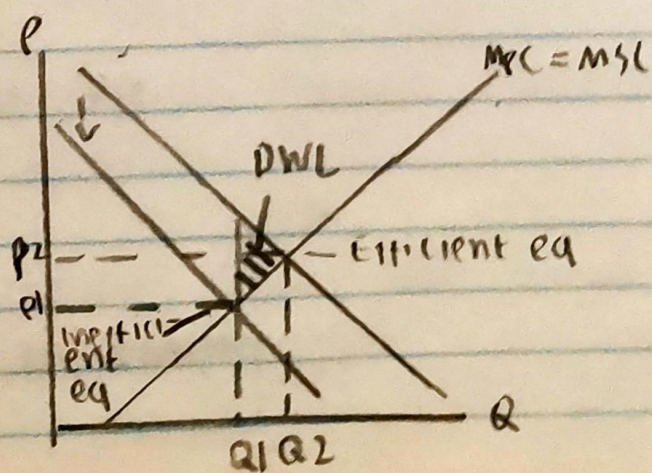
Fishing Negative production Externalities impact on allocative efficiency

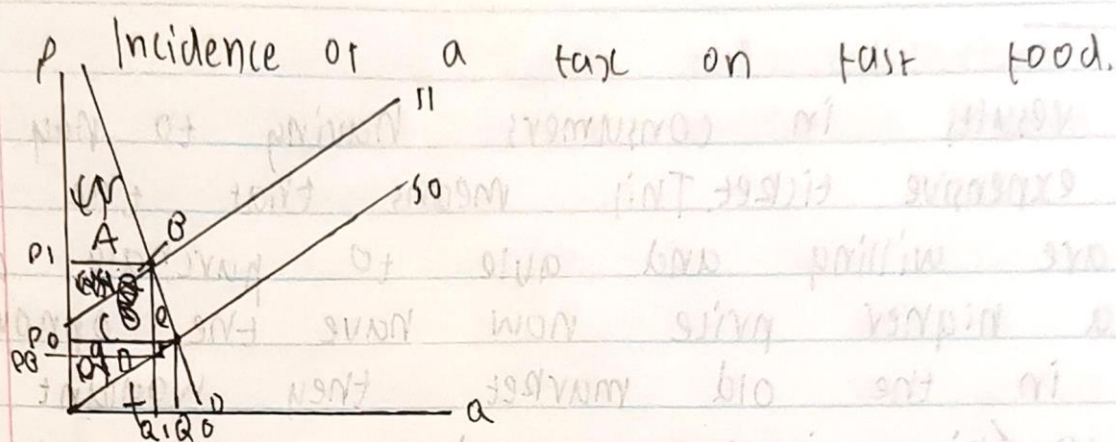


Market power impact on allocative efficiency



Healthcare positive consumption externalities and impact on allocative efficiency





In the diagram due to this tax there is a leftward shift of the supply curve ($S_0 - S_1$) resulting in a higher price ($P_0 - P_1$) as well as a lower quantity ($Q_0 - Q_1$).

Consumers now have to pay a higher price for the good. The consumer surplus is reduced which can be seen in the diagram as the original CS was AB while now it is only A due to the tax. Producers are also affected as due to this tax as producers receive a lower price (P_0) which means their total revenue goes down. (CDFK)

Producer surplus is also reduced due to TR going down ($CD - D$). Consumers bear a larger incidence of the tax seen in the difference between $P_0 - P_1$ in comparison to the producers incidence ($P_0 - P_0$).

This tax has created the DWL of E_f in the diagram due to resources not being allocated efficiently thus resulting in an inefficient market.

This also means that total welfare is reduced thus Variables/measures reducing equity.

- Unemployment
- Rate of inflation
- GDP

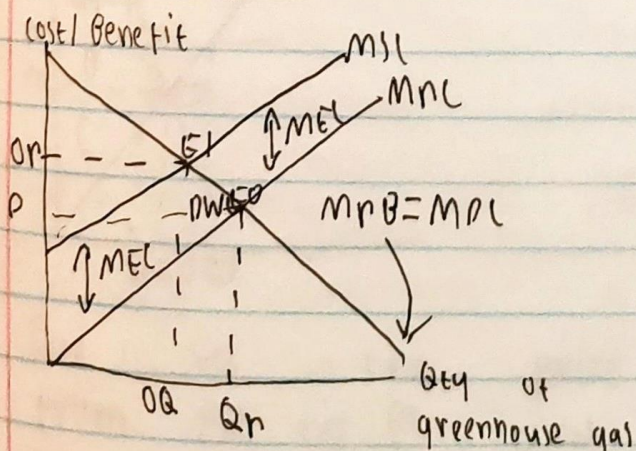
1a) Define the term negative externality.
 A negative externality is a negative cost incurred by a third party due to the production or consumption of a good or service.

1b) Greenhouse gas

1c) Common property resources are rival and non-excludable which means that anyone can harvest the resources but once one person consumes the good no one else can until it is replenished. The tragedy of commons occurs when a common property resource is overconsumed and is depleted because there is not enough time to replenish the resource.

ii) How can this problem be internalised?

A tax on greenhouse gases could be imposed on producers. The tax would force companies to pay for their negative external costs of producing greenhouse gases by increasing production costs. Companies should respond by reducing their production of goods that produce greenhouse



In the diagram Marginal social cost is larger than marginal private cost. This is due to the negative externality that the consumers incur from

the poorer air due to greenhouse gas production. In this case the equilibrium has shifted from E_0 to E_1 which has resulted in a higher price (O_p) and quantity (O_q) compared to ~~market eq.~~ the original market equilibrium. In the diagram there is a dead weight loss that occurs in the market as a reduction of welfare in the market due to an overproduction of the good arising from an overallocation of resources. The production of greenhouse gases creates the marginal social cost where marginal private cost is added to marginal external cost x which is the total cost incurred by society. The new equilibrium shows a higher socially optimal price and a lower socially optimal quantity supplied which reflects marginal social benefit being equal to the costs incurred by society. This would mean that producers would have to produce less of a good at a higher price to internalise the externality.

2018 Short answer

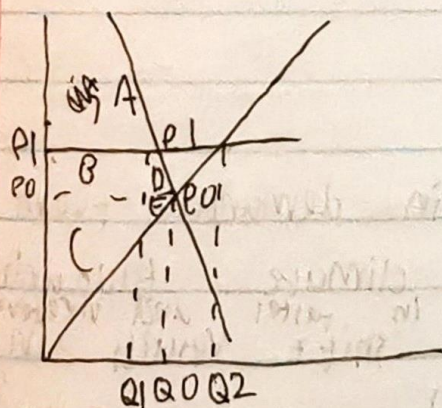
1. The supply curve slopes from left to right as at a higher price producers will want to produce more of a good to capitalise off of the higher prices as per the law of supply.

PED is a measure of the change in the quantity demanded of a product in relation to its price change.

$$\frac{-1}{2} = 0.5 \text{ PED}$$

Inelastic demand

The price elasticity of demand is important as it provides a basis for maximising profits. Since the potatoes are have inelastic demand as price increases, quantity demanded will decrease by a less than prop. amount. This means that as price increases total revenue will also increase. Producers can increase price to maximise profits.

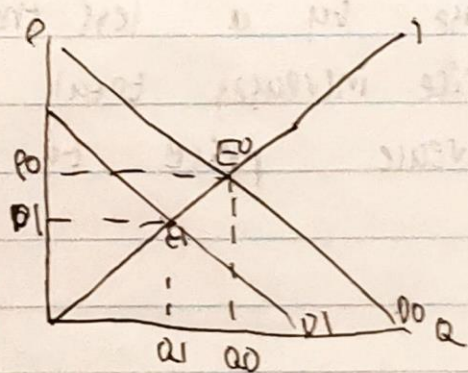


Due to the price floor, price is raised from P_0 to P_1 and quantity is reduced from Q_0 to Q_1 resulting in a new equilibrium at E_1 . This higher price has increased total revenue for producers thus resulting in an increase in producer surplus from C to BCx meaning

that producers have gained equity. As a result of the higher price and thus loss of purchasing power consumers have lost equity as seen in the fall of the consumer surplus from A to A' . This reduced purchasing power leading to reduced demand causes a surplus from Q_1 to Q_2 to occur. This loss of total welfare ^{and total well} as ~~surplus~~ results in a dead weight loss of DE which reflects an inefficiency in the market.

26 The lead time for an order is 1 year.

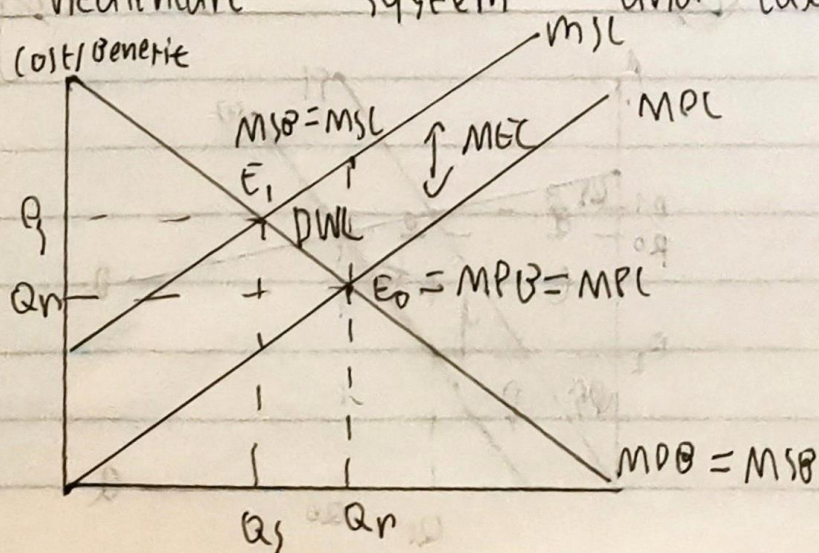
266 A product market is a market for final goods, for example a soccer ball.
A factor market is a market for the factors of production, for example cotton.



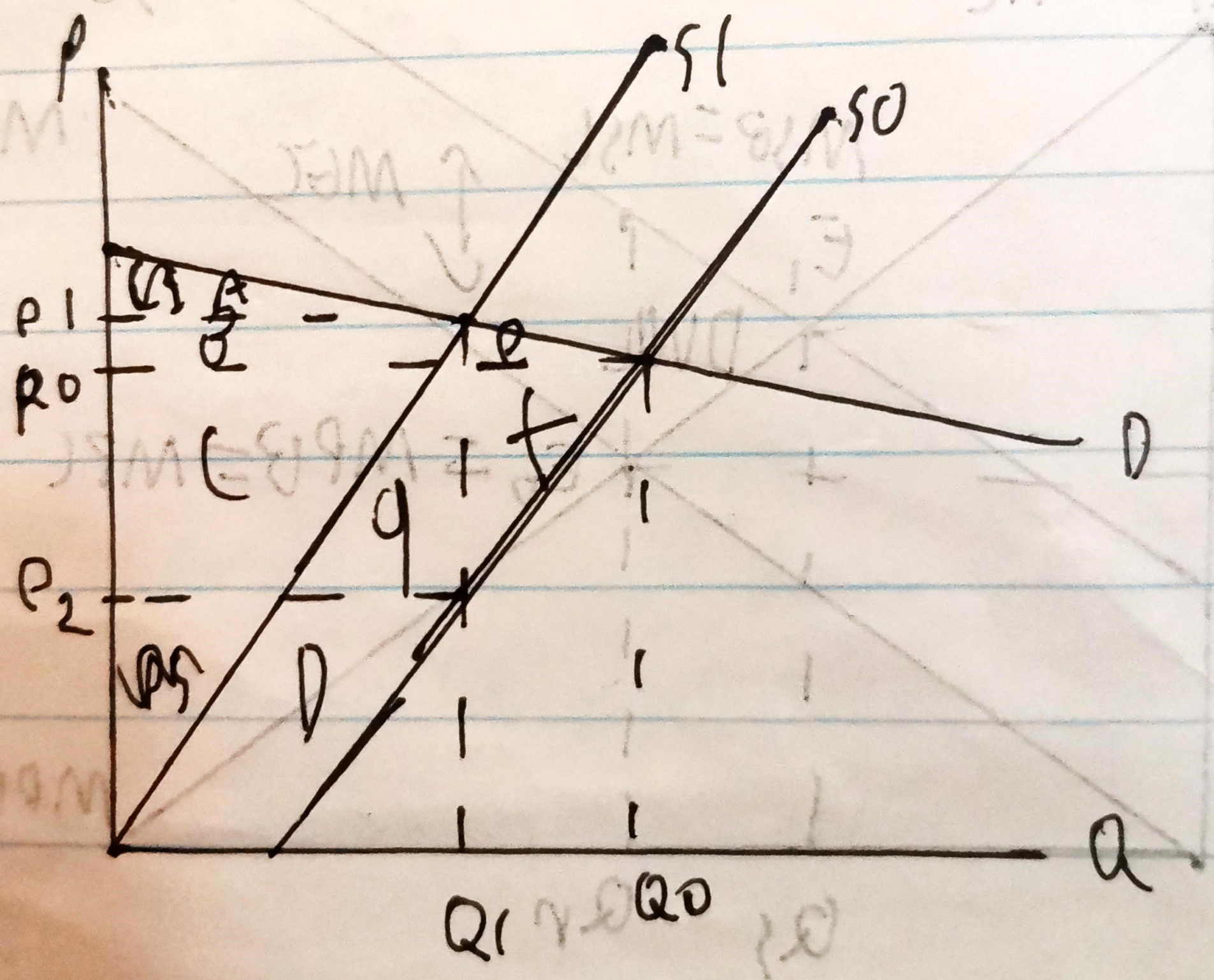
There is a shift left in demand from D_0 to D_1 due to the change in climate ^{move} ^{inelastically} ^{change in tastes and preferences} favouring summer clothes. This leftward shift results in a new equilibrium with a fall in quantity from Q_0 to Q_1 as well as a fall in price from P_0 to P_1 .

3 factors affecting demand for winter clothes. The weather is a factor as due to the weather becoming colder consumers would want to buy warmer winter clothes thus increasing demand. Consumer tastes and preferences are another factor affecting demand. As detailed in the excerpt, a longer summer resulted in consumers tending to be more inclined to summer clothes just as a result of this demand for winter clothes reduces. The level of disposable income would also affect demand for winter clothes as if disposable income was lower consumers might just use the same clothes from last year reducing demand. If consumer's disposable income was higher the quantity demanded for winter clothes would increase.

2A The consumption of soft drinks is a demerit good as the consumption of these goods consuming soft drinks is unhealthy for the body due to the higher quantities of processed sugars. Demerit goods have an external negative affect associated with their consumption. The consumption of soft drinks is a demerit good as it causes a negative externality due to obesity which uses up the healthcare system and taxpayer money.



In the diagram the market is producing at a level of Q_p . This ~~were~~ production level results in an external cost being incurred by consumers as seen by the marginal social cost curve being higher than the marginal private benefit curve. This gap between the curves is the marginal external costs incurred. At market equilibrium (E_1) marginal private benefit is equal to marginal private cost resulting in an inefficient equilibrium. In the diagram Q_s indicates a socially optimal output of resources which is a far less quantity supplied. This shift also results in a higher optimal social price at P_s . In the diagram a DWL occurs due to the overproduction ~~of resources~~ of the good arising from an overallocation of resources (Excess MSC over MSB at Q_p). This DWL results in a loss of total welfare indicating an inefficient market and thus market failure.

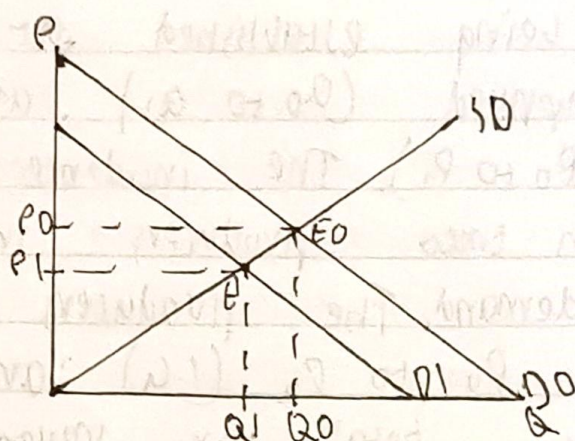


In the diagram as a result of the tax introduced there is a leftward shift in the supply curve from S_0 to S_1 . This reduction in supply results in a new equilibrium being established at E_1 with a reduction in quantity supplied (Q_0 to Q_1) as well as an increase in price (P_0 to P_1). The incidence of this tax is mainly placed on the producers in the diagram due to the elastic demand. The producers bear the burden of tax from P_0 to P_2 (CG) and producers bear P_0 to P_1 causing a total tax revenue of BCG. In the diagram without the tax at S_0 there was a consumer surplus (CS) of ABE and a producer surplus (PS) of CGFD. Once the tax was introduced (S_1) the CS was reduced to A while the PS was reduced to D. This reflects a total welfare loss and a loss of equity for consumers and producers. This loss of equity is reflected in the dead weight loss (DWL) that occurs at EF which represents thus indicating a lack of efficiency in the market.

2019

test

4 million $\times$ 2756 = 124 million revenue before
 2.5 $\times$ 4 million = 10 million revenue after

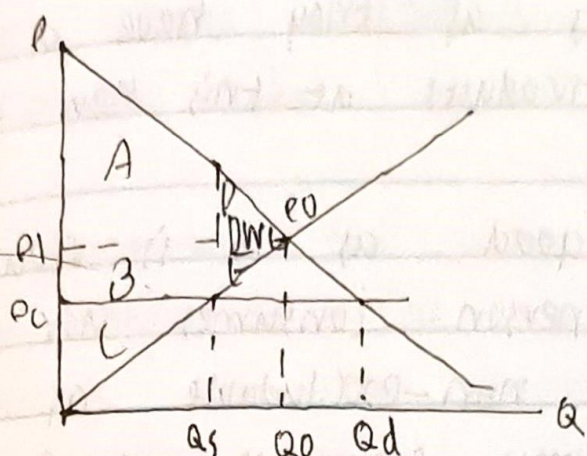


There is a leftward shift in the demand curve resulting in a fall in demand. This fall in demand results in a reduction in quantity from Q_0 to Q_1 as well as a fall in price from P_0 to P_1 . This causes a new equilibrium of Q_1 to be formed. This leftward shift in demand has likely been caused by the change in consumer tastes and preferences due to the needle crisis.

25) There is a high cost of production for strawberries which affects the supply of the product. The high price of strawberry crops makes it less accessible for farmers to grow. This could also result in a change of market size for strawberries which impacts the supply of the product. A change in market expectations such as the expectation that price will rise in the future would cause producers to hold ^{stock} supply until the price rises.

6a) An opportunity cost refers to the next best alternative forgone when purchasing a good. The opportunity cost forgone in this circumstance is government

expenditure on hospitals and education.



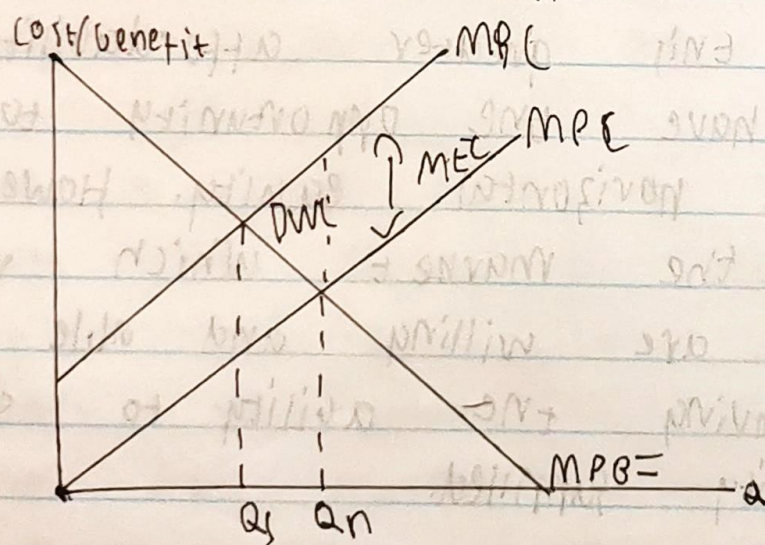
The lower price that occurs in the diagram results in the consumer surplus expanding from AD to AB while the producer surplus reduces from EBC to C. Overall there is a loss in total welfare as the expansion in CS is less than the reduction in PS. This results in a deadweight loss occurring at DE indicating an inefficiency in the market. Due to this lower ~~price~~ price consumers gain equity and have more opportunities to purchase homes with this greater affordability. Lower income levels now have the opportunity to purchase houses which increases horizontal equity. However, there is a shortage in the market which results in some consumers who are willing and able to pay a high price not having the ability to do so at the current quantity supplied.

266 A subsidy is an example of a government measure that promotes equity. A subsidy results in a rightward shift of the supply curve ~~and~~ ^{and a higher quantity supplied} ~~which~~ ^{thus} creating a lower price. This market is more accessible for lower income brackets to purchase thus increasing horizontal equity. A price ceiling is another measure that promotes consumer equity. A price ceiling directly creates a lower price for the consumers, but also a lower

quantity supplied. This lower price directly impacts lower socioeconomic classes as they have a far greater ability to purchase products at this new lower price.

Atmosphere is a public good as it is rival in consumption as when one person consumes air it degrades breathable air. It is also non-excludable as it is extremely costly and almost impossible to exclude people from consuming this good.

Since the goods are non-excludable it means that they are likely to be overconsumed as well as being rivalrous which results in the degradation of the resource with no compensation being paid. This directly impacts the air quality which can give a lot of people lung diseases such as asthma which puts pressure on the healthcare system.

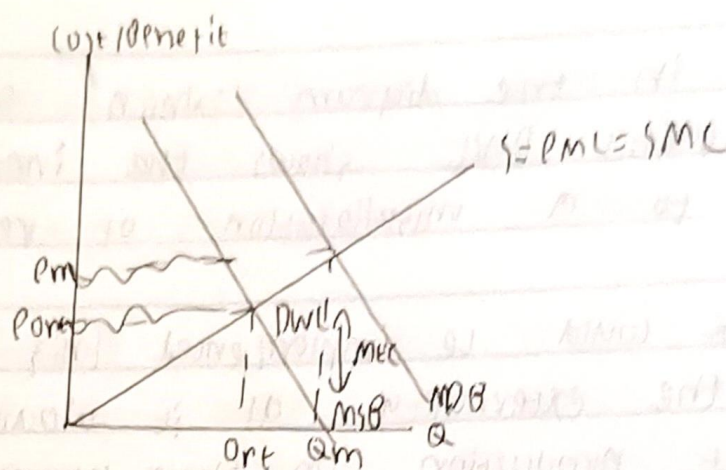


In this diagram there is a marginal social cost that is higher than marginal private cost which results in a marginal external cost. In the market this marginal external cost is not accounted for thus resulting in the gap. There is an overproduction of resources in the market as seen in the difference between quantity produced and socially optimal level of production. This overproduction of resources indicates overallocation.

tion of resources in the diagram which causes a dead weight loss. This DWL shows the inefficiency in the market due to a misallocation of resources.

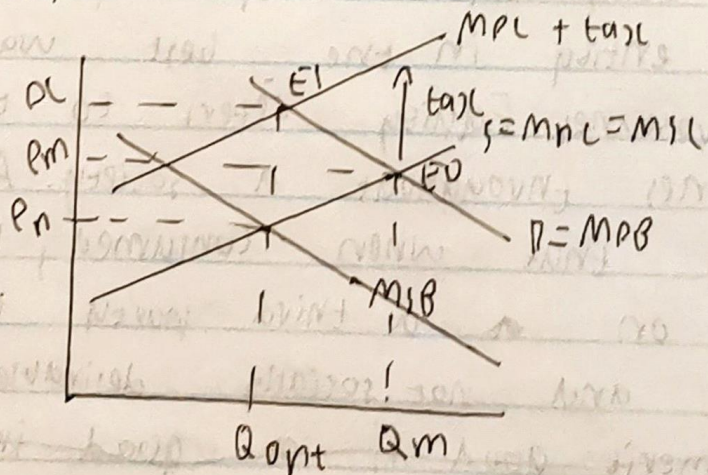
A Pigouvian tax could be implemented. This tax would help internalise the externality as it would raise the producers cost of production on these harmful goods, thus making them less inclined to produce them. This would make the quantity supplied go to a level closer to the socially optimal level, resulting in the negative third party impact ^{from} producing this good to become smaller. Regulations on producing harmful goods would have similar impact as a Pigouvian tax. It would internalise the externality as producers won't legally be able to produce more than a socially optimal level. This would also result in the negative third party impact becoming smaller as less of the harmful good and the detriment that comes from it are being produced.

Efficiency can be defined as an economic state in which resources are being allocated optimally to serve each individual or entity in the best way while also not wasting resources. Equity refers to the equal distribution of resources throughout a society. A demerit good is a good that when consumed, has a negative impact on a third party due to the ~~unhealthy~~ unhealthy and not socially desirable nature of these products. ~~A merit good is a good in which~~
~~when~~ ~~that~~ ~~when~~ consumed has a



A demerit good is a good that creates a negative third party impact. In the diagram above the marginal private benefit is higher than the marginal social benefit. This divergence between MPB and MSB reflects the marginal external cost seen when consuming demerit goods that impacts the third party. In the diagram the market is producing at Q_m which is far higher than the optimal consumption of Q_{opt} . This leads to a loss of total welfare thus creating a DWL from the over consumption leading to the overallocation of resources in the market. This is a failure in the free market.

A way to internalise costs is to tax demerit goods.



In the diagram above the tax implemented causes a leftward shift in the supply curve. This leftward shift meets with the marginal private benefit curve, to create

a new equilibrium at e_1 . This equilibrium has a higher price of P_c which results in a contraction of demand. This contraction of demand leads to a reduction in quantity supplied which has brought it back to a socially optimal level. Overall a tax is an optimal way to internalise the negative externality from a demerit good. As the higher price it creates causes a contraction in the quantity demanded thus bringing quantity supplied to and a socially optimal level. This reduces the marginal external cost and the third party impact as less people are consuming these demerit goods.

Macroeconomic indicators

- Real GDP (National income)
- Unemployment rate
- Inflation rate (General price level)
- Government budget
- Balance of payments and exchange rates
- Sustainable economic growth 3.4%
- Price stability low inflation 2.3%
- Full employment low unemployment 5% - 0% cyclical
- Exchange rate stability or favourable balance of payments
- Inflation - reduces purchase power of consumers, reduces saving incentive, exports are more expensive while imports are cheaper

Inflation - sustained (inordinate) increase in general price level
10% to 7% inflation - decrease in rate of inflation - disinflation

5% - 7% increase in rate of inflation

(CPI) - consumer price index measure of the cost of living or the cost of goods and services purchased by the typical household of an economy.